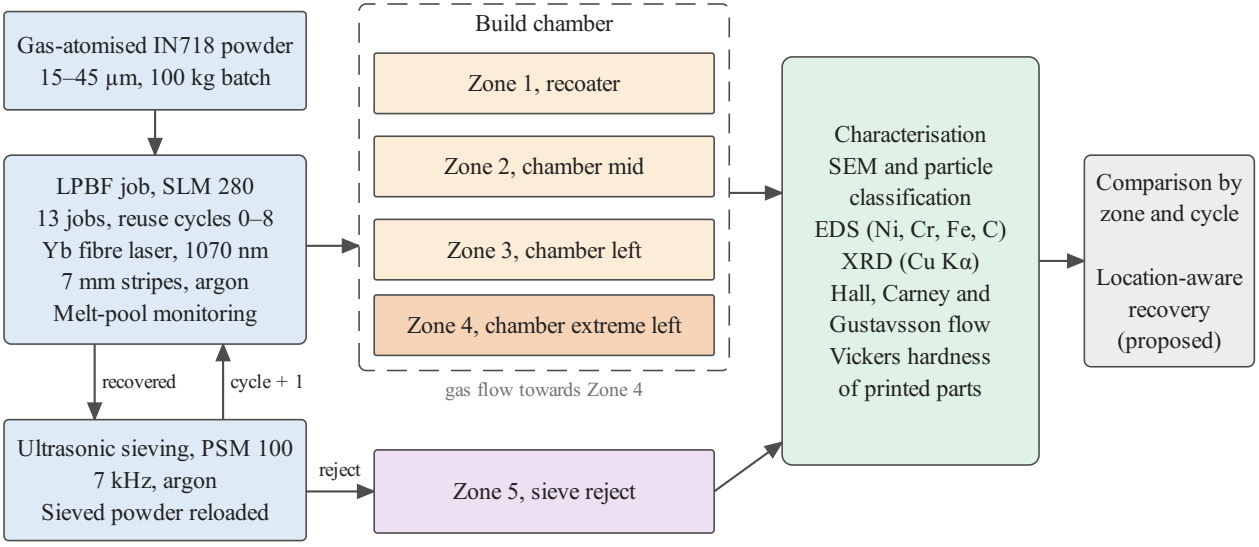
Gas-atomised IN718 powder
15–45 µm, 100 kg batch
LPBF job, SLM 280
13 jobs, reuse cycles 0–8
Yb fibre laser, 1070 nm
7 mm stripes, argon
Melt-pool monitoring
recovered
cycle + 1
Ultrasonic sieving, PSM 100
7 kHz, argon
Sieved powder reloaded
reject
Build chamber
Zone 1, recoater
Zone 2, chamber mid
Zone 3, chamber left
Zone 4, chamber extreme left
gas flow towards Zone 4
Zone 5, sieve reject
Characterisation
SEM and particle classification
EDS (Ni, Cr, Fe, C)
XRD (Cu Kα)
Hall, Carney and Gustavsson flow
Vickers hardness of printed parts
Comparison by zone and cycle
Location-aware recovery (proposed)